Abstract

Urban green spaces are critical for maintaining biodiversity and supporting wildlife. In temperate regions, resident birds face significant challenges during winter, some of which may be exacerbated or, conversely, alleviated in urban settings. To address those challenges or opportunities, even "resident" birds may complete very short-distance, within-city migrations to take advantage of seasonal habitats. To increase our understanding of non-breeding habitat use of resident birds in temperate urban areas, we experimentally translocated Northern Cardinals (*Cardinalis cardinalis*) among sites spanning an urbanization gradient in Montreal, Canada. We monitored individual movements throughout the non-breeding season using VHF telemetry to quantify home range size, habitat selection across spatial scales, and landscape habitat associations. Cardinal populations on Christmas Bird Counts increased across the northeastern United States and southeastern Canada in the past 100 years, particularly in cities, potentially due to anthropogenic food resources (e.g., bird feeders). While cardinal core home range sizes during the non-breeding period were similar across sites, the total home range was significantly larger at urban sites compared to peri-urban sites. Model fit was best at the smallest scale across all site types. Habitat associations differed across site types, except for proximity to bird feeders, which was consistently related to habitat use by cardinals. In peri-urban sites, cardinal use was significantly greater at sites with lower vegetation and greater vertical complexity, suggesting a preference for heterogeneous or shrubby vegetation. We highlight the importance of fine-scale ecological understanding to guide year-round green space management for urban-adapted species.

Keywords: behavioral flexibility, fine-scale movement, habitat association, home range size, manual VHF telemetry, non-breeding period, scale of habitat selection, urbanization gradient

Lay summary

- Urban green spaces can help wildlife survive in cities, but little is known about how birds use these spaces during the winter months in northern climates.
- We tracked Northern Cardinals in Montreal, Canada, using small radio transmitters to document movements and discover the habitats used
- Cardinals spending the winter in the most built-up areas used larger territories and were most often in areas with bird feeders and lots of roads, such as residential areas
- Cardinals show flexibility in how they use habitat depending on how urban it is, suggesting well-designed urban green spaces can support urban-adapted birds year-round.

Introduction

Urbanization is accelerating worldwide (Wang et al. 2012; Elmqvist et al. 2013; Simkin et al. 2022), leading to habitat fragmentation, habitat loss, and chronic pollution in and near urban landscapes (Lowry et al. 2013; Seress and Liker 2015; Murray et al. 2019; Vaz et al. 2023). Future global urban expansion could lead to an additional 11-33 million hectares of habitat loss by 2100, with a disproportionately large increase in habitat fragmentation (Li et al. 2022). Urbanization in turn reshapes wildlife spatial ecology, habitat use, and may offer anthropogenic resources that species with behavioural plasticity may exploit for enhanced fitness opportunities. Urban landscapes, and particularly anthropogenic features such as buildings, roads, and other impervious surfaces, may restrict or alter animal movement or habitat use. This may in turn lead to different patterns of home range sizes and movement patterns for individuals of the same

species within vs. outside of urban areas (O'Donnell and delBarco-Trillo 2020). An extensive meta-analysis found that several small bird and mammal species occupied smaller home ranges, with restricted spatial movement, in urban areas (O'Donnell and delBarco-Trillo 2020; but see opposite trend for predators: Riley et al. 2003; Thompson et al. 2021; Hindmarch et al. 2017). Smaller home ranges may be a result of locally abundant and reliable resources such as food or shelter, diminishing the need for extensive movement (Rutz 2006; Bino et al. 2010; Šálek et al. 2015; Reher et al. 2016; Mitsuhashi et al. 2018), or a high density of conspecific competitors forcing limited territory use (Galeotti 1994; Boggie and Mannan 2014). Additional ecological drivers beyond resource use may influence home range sizes. For example, it was suggested that for *Zonotrichia leucophrys* (White-crowned Sparrows) in the San Francisco Bay area, limitations of communication distance in a noisy urban landscape may drive individuals into smaller home ranges during the breeding season (Simberloff et al. 2023).

An individual's potential to respond to these urban pressures through behavioural adjustments ultimately depends on the species' behavioural flexibility. Urban areas hold both novel challenges and opportunities, including increased predation pressure, higher disturbance levels, novel anthropogenic resources, and human-made structures. To compensate for disturbances from noise, light, and human activity, some species may alter their nesting behaviour, including nesting higher in the canopy (e.g., *Pica pica* (Eurasian Magpie); Šálek et al. 2020) or using off-ground artificial surfaces, leading to increased fitness (e.g., *Junco hyemalis* (Dark-eyed Junco); Bressler et al., 2020). Urban habitats also provide foraging opportunities which can be exploited by some species like *Falco peregrinus* (Peregrine Falcon), which uses buildings as perches and nesting sites, capitalizing on disoriented migratory birds attracted to lit glass structures (Kauffman et al. 2003; DeCandido and Allen 2006; Altwegg et al. 2014).

Similar to many wildlife studies (Marra et al. 2015), urban ecology research is strongly biased toward the breeding season. This is problematic because birds often exhibit greater urban tolerance and behavioural flexibility during the non-breeding season (Callaghan et al. 2021). Urban environments may also provide important winter refuges through predictable anthropogenic food sources and the urban heat island effect (Jokimäki et al. 2022; Latimer and Zuckerberg 2021). Winter occupancy dynamics are highly species-specific and influenced by thermal tolerance, particularly at northern latitudes where energetic demands increase substantially (Fristoe 2015). For example, *Cardinalis cardinalis* (Northern Cardinals) form winter flocks and alter their space use outside the breeding season (Ritchison and Omer 1990). Birds overwintering at northern latitudes, such as southern Quebec, are expected to spend considerably more energy maintaining body temperature than those wintering farther south, which is especially challenging given the scarcity of food resources during the cold months. Despite these challenges, winter remains underrepresented in ecological research (Dinh et al. 2023), highlighting the need to better understand how birds use urban habitats during this critical season.

To address the gap in knowledge about winter-resident urban birds, we studied *Cardinalis cardinalis* across an urban gradient in one of Canada's largest cities, Montreal, Quebec. Cardinals are medium-sized songbirds (weighing 40 to 45 grams), capable of supporting radio tags lasting almost the entirety of the non-breeding season. Over the last 100 years, cardinals have expanded their distribution northward, especially in association with urban and suburban landscapes, and now occur in regions of extreme cold and limited winter resources, such as southern Quebec (Beddall 1963; Smith et al. 2011; Robert et al. 2019). As a translocation experiment, we captured and released individual cardinals at sites distributed along an urban

gradient to broadly quantify habitat use and movement dynamics as a function of urbanization during the non-breeding season. Our study had four objectives: (i) to investigate cardinals' home range size as a function of urbanization, (ii) to quantify the spatial scale of habitat use associations, (iii) to determine possible predictors of cardinal habitat use from anthropogenic and natural habitat variables, and (iv) to qualitatively document the spring movements from non-breeding to breeding territories. We predicted that wintering cardinals require larger home ranges and select habitats at larger scales when in dense urban locations due to food sources being more scattered than in less urban areas (suburban/peri-urban; (Havlíček et al. 2022). We also predicted cardinal habitat use will be positively associated with food resources (bird feeders), preferred vegetation structure (heterogeneous vertical vegetation structure), and lower human population density. Finally, we predicted that cardinals would transition from their breeding sites in forested/canopy covered areas towards residential areas in the fall to take advantage of bird feeders during winter and reverse this localized migration in spring.

Materials and methods

Study locations

The island of Montreal is located in southern Quebec, Canada, with a population of ~2 million people, including areas of high population density. The local climate comprises four distinct seasons with cold winters (temperatures can range between -23 ˚C and 4 ˚C during the month of January) and warm summers (with temperatures sometimes exceeding 30 ˚C; WeatherSpark). We conducted our research at three distinct study sites (Figure 1) reflecting different levels of urbanization. Each site includes two release locations (150 m apart), randomly assigned to each individual. The peri-urban site is located on the western end of the island of Montreal at the McGill Bird Observatory (MBO; 45˚25'51.4"N 73˚56'18.5"W). The MBO is part of the 30 km^2

large urban park named *Grand parc de l'Ouest*. The suburban site is located in the Town of Baie d'Urfé (45˚24'39.2"N 73˚55'50.0"W). The urban study site is located in the Notre-Dame-de-Grâce neighbourhood of the city of Montreal(45˚27'35.7"N 73˚38'30.2"W). Cardinals are regularly present at all three sites, supporting our assumption that translocated birds should be able to survive the winter at each location.

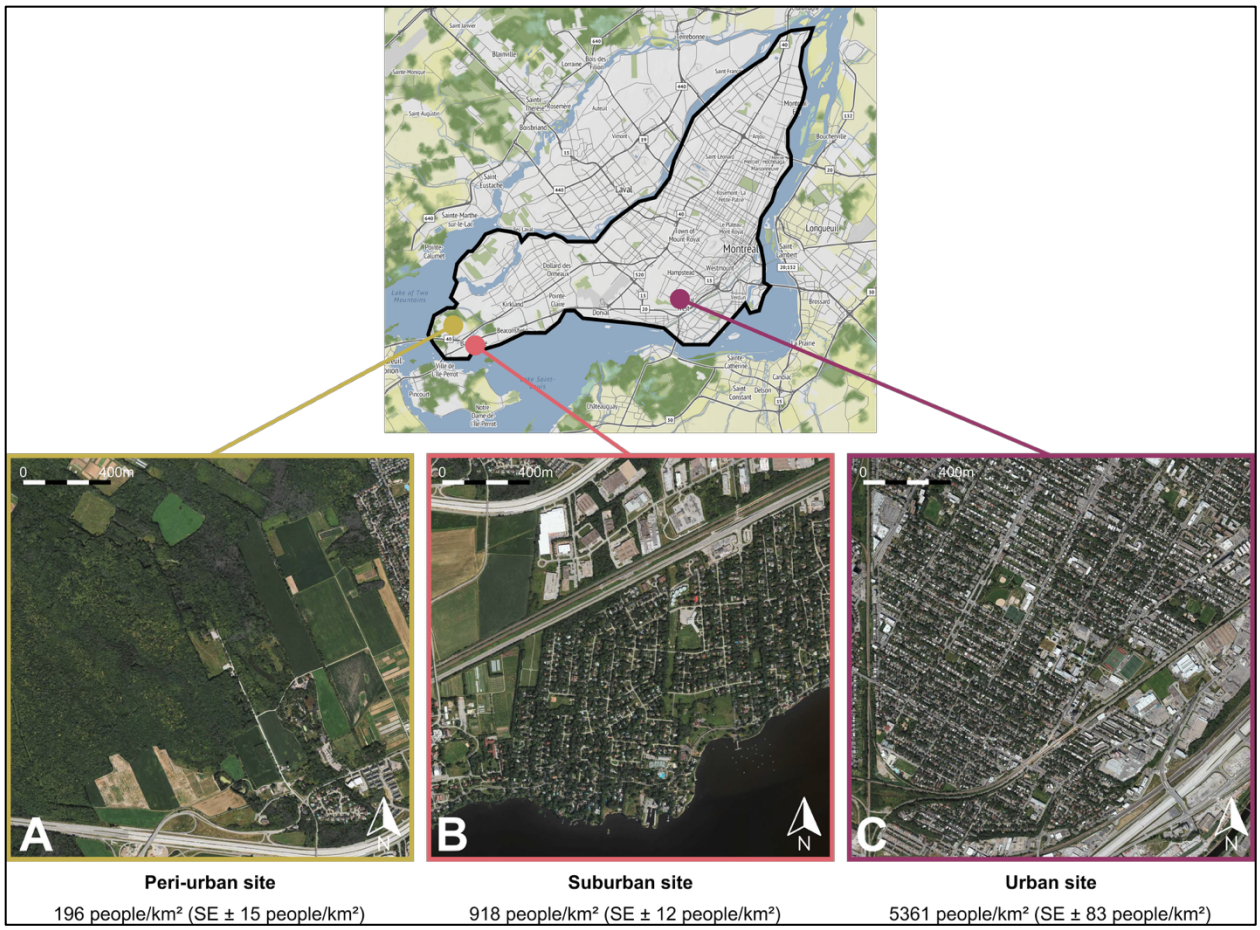


Figure 1. Study locations on the Island of Montreal A) The peri-urban site is at the McGill Bird Observatory (2% of the area covered by streets and/or roads). B) The suburban site (5% of the area covered by streets and/or roads) is located in the residential Town of Baie d'Urfé. C) The urban site (13% of the area covered by streets and/or roads) is located in the Notre-Dame-de-Grâce neighbourhood.

The three translocation sites differed significantly in their level of urbanization, based on human population density data from the Cancensus Program R package (von Bergmann et al. 2021), using census data from the Government of Canada for the year 2021 (Kruskal-Wallis test and a Bonferroni-corrected pairwise Wilcoxon test, $p < 0.001$). All three sites are characterized by relatively high levels of green space compared to the median canopy cover for the Montreal area (~ 26%) (HealthyDesign.City 2021). As such, our gradient does not fully capture the extent of urbanization present across the entire city and should be interpreted as a localized urbanization gradient. We initially translocated a few cardinals to a highly urban area with lower canopy cover in the city core, but due to two immediate mortalities following translocation, we decided to omit that extreme of the urban gradient from our study.

Radio tagging, banding & translocation

All birds were caught during routine fall migration monitoring at MBO and were fitted with colour bands and Motus NanoTags following standard banding and morphological measurements (see Supplemental Information). Cardinals translocated to sites away from the peri-urban location (suburban and urban), as well as control birds held at the peri-urban site, underwent a holding time of 45 minutes before release to standardize treatment amongst individuals. For the winter of 2022-2023 (winter 1), we were able to consistently track 20 birds released as controls at the peri-urban site, which served as a pilot winter to assess the feasibility of the project. During the winter of 2023-2024 (winter 2), we tagged and banded 45 birds, with 15 individuals translocated to the urban site, 15 translocated to the suburban site, and 15 released as controls at the peri-urban site. In 2024-2025 (winter 3), we tagged and banded 42 birds, with 5 individuals released as controls at the peri-urban site, 16 individuals translocated to the suburban site, and 21 individuals translocated to the urban site.

147 Although tags used in this study were lighter than those used in previous work on cardinals, we
148 considered the potential for tag effects when interpreting our results. Barron et al. (2013) showed
149 that during the reproductive season, cardinals with tags (weighing 1.6 g [double the 0.8 g of the
150 tags used in the present study]) defended their nests less vigorously, but did not change their
151 provisioning effects. Cardinal nests with tagged parents also had lower daily survival rates and
152 were less likely to fledge offspring (Barron et al. 2013). However, these tag effects were
153 relatively modest and occurred across diverse habitats, making it unlikely that they would
154 manifest only in our dense urban site while remaining undetected in peri-urban and suburban
155 locations. Birds were banded under Animal Use Protocol 2007-5447 from McGill University and
156 federal banding permits 10973, 10973B, 10973D issued by the Canadian Wildlife Service, and
157 radio-tagged under Animal Use Protocol 22BF01 and 23BF01 from Environment and Climate
158 Change Canada.

159 Cardinal abundance and seasonal distribution

160 We used the number of cardinals recorded per party-hour in Audubon Christmas Bird Count data
161 between 1900 and 2015 to compare historical cardinal abundance in Quebec, Canada, with
162 Maine, New York, New Hampshire, and Vermont (United States) (National Audubon Society
163 2020).

164 Non-breeding season tracking and home range estimations

165 We monitored tagged birds throughout the non-breeding season using an array of automated
166 Motus towers and handheld telemetry receivers. We conducted manual radio tracking of each
167 individual two to three times per week between the months of October and May each year. The
168 areas monitored were ~5 km^2 (500 ha) at the peri-urban site, ~6 km^2 (600 ha) for the suburban
169 site, and ~18 km^2 (1800 ha) for the urban site. Tracking effort and spatial coverage differed

among sites because weather conditions and landscape structure imposed logistical constraints. At the peri-urban site, the lack of road networks and lower signal interference allowed the tracking team to move more freely and locate individuals more quickly, resulting in a smaller monitored area. In contrast, tracking the birds in suburban and urban areas meant that the tracking team's movements were constrained by road access, built infrastructure, private land, and higher levels of radio interference. In addition, individuals released at the suburban and urban sites often moved farther away from their release location, requiring the tracking team to search across larger areas to maintain contact with tagged birds. These differences in spatial coverage reflect practical limitations of tracking birds in the heterogeneous landscape of urban environments and should be considered when comparing movement patterns across sites. The people tracking the birds by hand (trackers) aimed to record at least one GPS location for each individual per week using a Garmin ETREX 22x GPS. Trackers typically spent 5 hours tracking per session, twice per week, and started by first searching for newly relocated birds and/or birds with known recent locations. Once these individuals were found, trackers moved on to expand their search areas to find missing individuals. Details on the tracking method and estimations of precision can be found in the Supplemental Information section.

For all three study sites, we calculated the average home range size using the Kernel Density Estimation (KDE). The home range derived from the utilization distribution (UD) represents the minimum area containing a specified proportion of an individual's space use (i.e., the GPS location recorded in the field) with the kernelUD function from the adehabitatHR: Home Range Estimation R package version 0.4.22 (Worton 1989; Calenge 2023, 2024). For example, the 95% home range represents the smallest area in which there is a 95% probability of finding the individual. We selected the 95% and 50% contours to investigate, respectively, the core area used

193 within the home range of the birds and the broader home range excluding the most extreme
194 locations (outliers). A minimum of 5 GPS locations were required to adequately run the function
195 to calculate the KDEs, meaning that individuals with fewer points than this threshold were
196 excluded.

197 Resource Selection Function

198 We compared 'used' vs. 'available' habitat using a resource selection function (RSF) for
199 cardinals during the non-breeding season across our three sites. We used environmental and
200 anthropogenic variables extracted from available spatial data in addition to field-collected data.
201 This included mean and standard deviation (SD) of vegetation height (1m resolution) from
202 OpenForest (Government and Municipalities of Quebec 2016), distance to the nearest bird feeder
203 (collected through systematic field observations), human population density data (census unit)
204 using the Cancensus R package (von Bergmann et al. 2021), and spatial data on the streets and
205 roads from the island of Montreal (Ville de Montréal 2018). Unused locations were derived at a
206 1:1 ratio by creating random locations within 300 meters of the used location, at a random angle.
207 We used the matched case-control design elaborated by Fortin (2005, 2009) and applied a 300 m
208 buffer to reflect the median daily bird movement. Our analytical design (population-level RSF)
209 included individual IDs as random effects and evaluated all used-available pairs from all
210 individuals together. As such, we did not require a minimum number of locations per individual,
211 as we would for an individual-level RSF. All predictors were scaled and centered prior to
212 analyses.
213 We built our candidate model set to compare the relative strengths of anthropogenic and natural
214 vegetation variables to predict cardinal habitat use. This included four *a priori* hypothesis-based
215 models including, (1) the "Full model" (Used ~ Mean vegetation height + Sd vegetation height +

Population density + Road length + Distance to nearest feeder + (1|Winter) + (1|Site) + (1|ID)), including all of the variables, plus the winter (years of sampling), relocation site and bird ID as random effects to account for the additive importance of anthropogenic and natural vegetation variables, (2) the "Vegetation height" model (Used ~ Mean vegetation height + (1|Winter) + (1|Site) + (1|ID)), assuming the vertical height of vegetation was the dominant variable related to habitat use by cardinals, (3) the "Sd vegetation height" model (Used ~ Sd vegetation height + (1|Winter) + (1|Site) + (1|ID)), assuming that cardinal habitat use is related to vertical vegetation composition, and (4) the "Anthropogenic" model (Used ~ Population density + Road length + Distance to nearest feeders + (1|Winter) + (1|Site) + (1|ID)), which assumed that cardinal habitat use would be related to anthropogenic variable We also included a "Null" model, with random effects only (Used ~ + (1|Winter) + (1|Site) + (1|ID)). We ranked all four models according to the corrected Akaike Information Criterion (AICc) following the method previously described by Burnham and Anderson (2002). We used Variance Inflation Factor (VIF) to test for multicollinearity among predictor variables (variables with VIF values > 10 were considered to have problematic collinearity and were removed) and Hosmer-Lemeshow as a goodness-of-fit test (Hosmer and Lemesbow 1980). VIFs ranged between 1.05 and 3.31 across all predictor variables; thus, none were removed. All statistical analyses were performed using Program R version 4.4.2 (R Core Team 2024).

Scale-of-effect

We used a data-driven approach to identify the pertinent spatial scale at which variables have the strongest relationship with a species' response (i.e., scale-of-effect, Martin & Fahrig, 2012) by extracting variables at four spatial scales (25, 50, 100, 200 meters). We chose these spatial scales as they spanned a range that was small enough to capture finer responses as allowed by the

spatial grain of the vegetation data, but also large enough to capture daily movement of tracked birds. For each scale, we ran generalized linear mixed models (GLMMs) separately for each site type (peri-urban, suburban, urban) using all five predictor variables as well as year, release location, and individual ID as random effects. We ranked models using Akaike Information Criterion (AICc) values, which take into account small sample sizes (Burnham and Anderson 2002). The best model was identified as having the lowest AICc value. We also calculated the ΔAICc to evaluate competing models following Burnham and Anderson (2002) (models with a ΔAICc < 2 were considered to have substantial support, competing with the best model and models with a ΔAICc > 10 were deemed to have no support, not competing with the best model). We acknowledge that low per-individual sample sizes might reduce the precision of the scale-of-effect estimates, as models fitted on few observations per individual could be susceptible to individual-level variation driving apparent scale preferences. To mitigate this, we included individuals' IDs as a random effect and interpret scale-of-effect results at the population level rather than as individual-level habitat selection.

Because the four buffer sizes represent nested spatial extents, we expect the variables extracted from these buffers to be positively correlated. We assessed cross-scale correlations for each predictor type using Pearson's "r". Correlations were highest for human population density (r = 0.96-0.99 across all sites), moderate for mean vegetation heights (r = 0.37-0.94) and road length (r = 0.32-0.95), and lowest for SD vegetation height (r = 0.33-0.84), decreasing with increasing scale separation in all cases. Variables from different scales were never included within the same model—each candidate model uses a single buffer size—so cross-scale correlations do not inflate within-model multicollinearity (all VIF < 10). The buffer comparison therefore constitutes a scale-selection exercise rather than a multi-scale additive approach. Once the scale-of-effect

was determined for each location, we conducted a resource selection function at the site-type-specific spatial scale.

Results

Historical and seasonal context of Cardinals in southern Quebec

To place our winter telemetry results in the context of the species' recent expansion into southern Quebec and its seasonal dynamics within the study region, we summarized long-term regional abundance from Christmas Bird Count (CBC) data and local bird banding records. In southern Quebec, cardinals have increased in abundance since the 1970s as documented in the CBC, reflecting the species' well-documented northward range expansion from the northeastern United States into southern Quebec beginning in the 1960s (Figure 2).

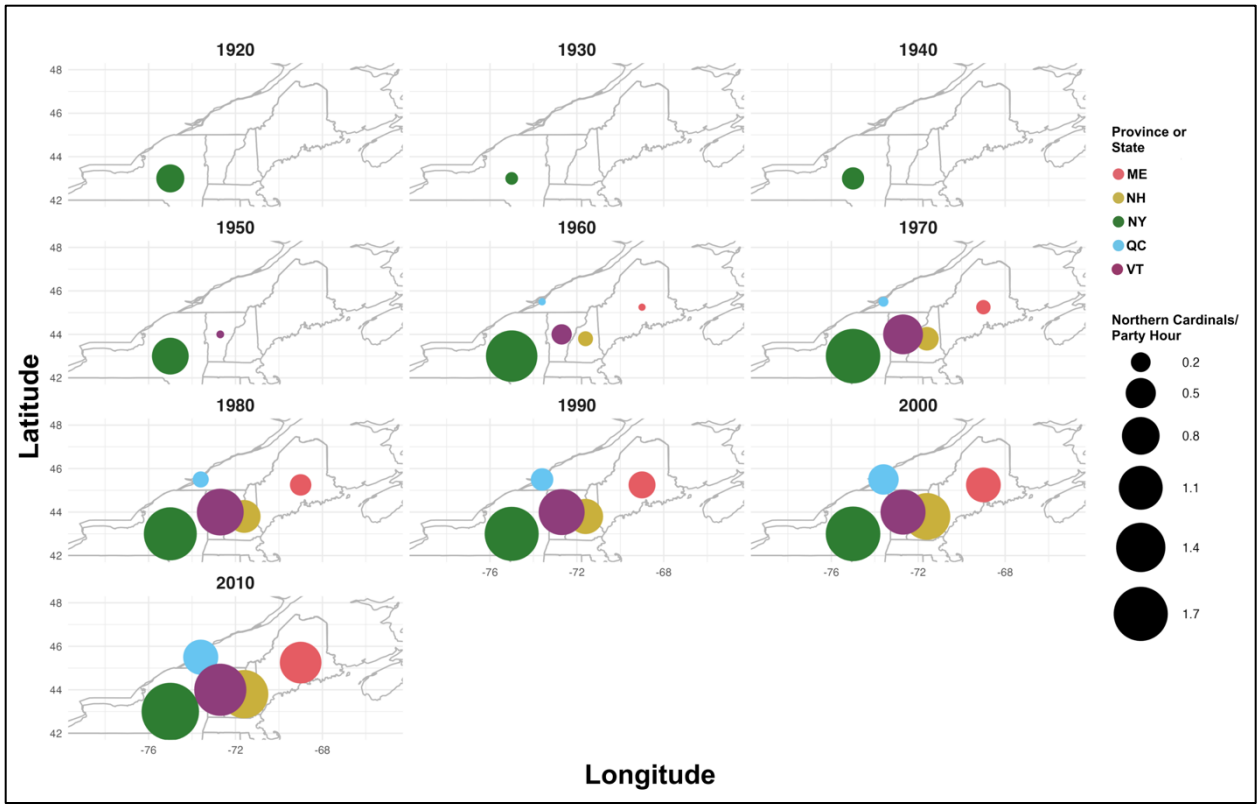


Figure 2. Number of cardinal observations per party hour between the 1920s and the 2010s according to state (United States) or province (Canada) as part of the Audubon Christmas Bird Count (CBC).

Nineteen years of banding data from the McGill Bird Observatory also revealed distinct seasonal patterns in captures (Figure 3). Capture rates were higher during the fall monitoring program (~0.75 birds/h) than during the spring program (~0.27 birds/h), accompanied by a greater proportion of unbanded individuals, indicating an influx of new birds in late fall. Although most fall captures were juveniles or individuals of unknown age (Figure S1B, C), adults also exhibited peaks in mid-May and late July (Figure S1A), suggesting that both post-fledging dispersal and local movements between breeding and wintering areas contribute to seasonal variation in captures.

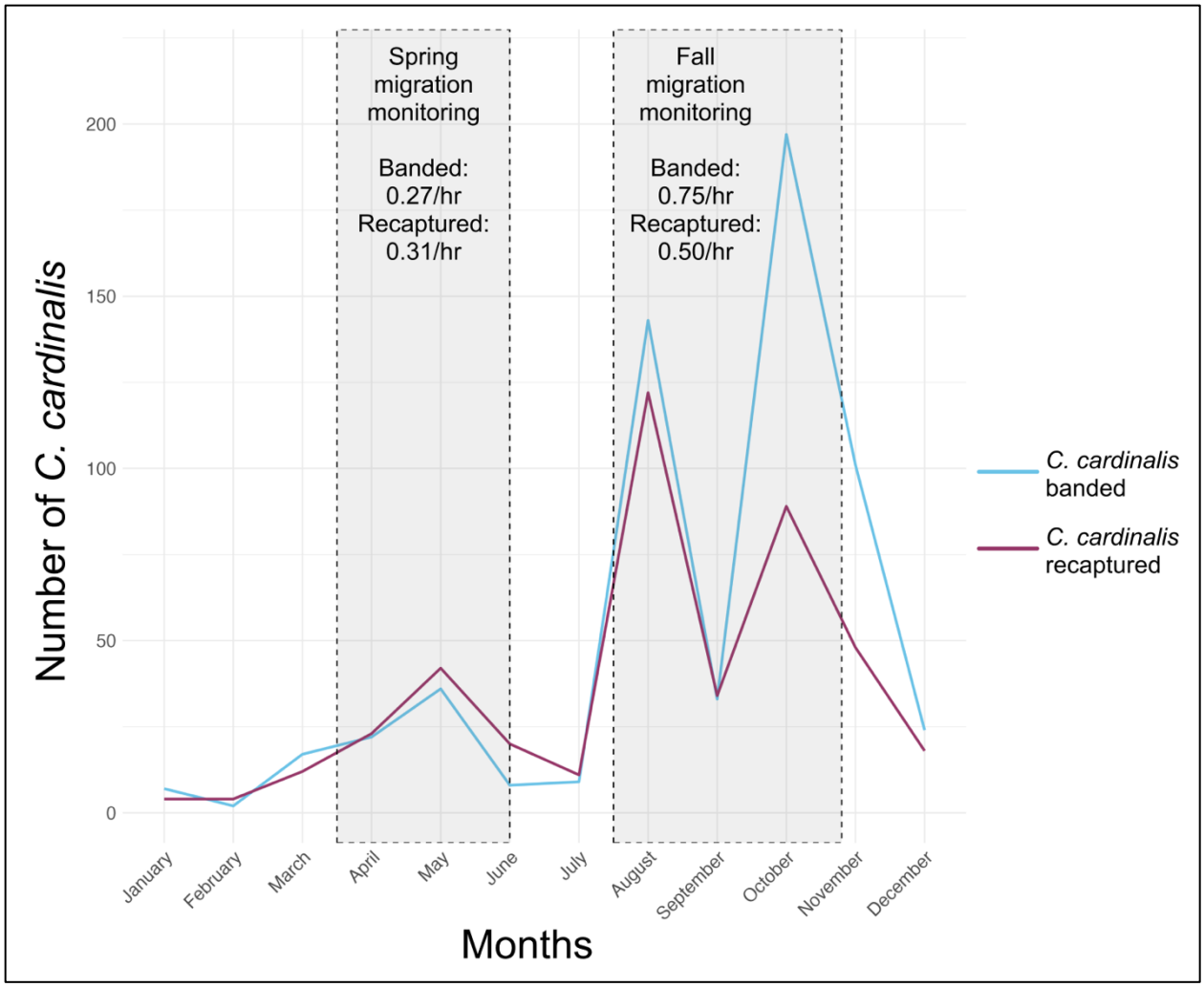


Figure 3. Number of cardinals banded and recaptured at the McGill Bird Observatory (peri-urban site) between 2004 and 2022 in Montreal, Quebec, Canada. The grey boxes highlight the duration of the spring and fall migration monitoring programs, along with the approximate number of cardinals banded and recaptured per hour of mist nets being opened during the season.

Telemetry tracking of cardinals across an urban gradient

Telemetry tracking was conducted 2 to 3 days per week between October and May 2022-2024, for a total of ~288 days of monitoring per winter (Table 1). We recorded a total of 688 unique detections at the peri-urban site, 411 at the suburban site, and 265 at the urban site across the

three years. The proportion of cardinals tracked one month post-release differed significantly among sites (Pearson's $\chi^2 = 21.85$, df = 2, $p < 0.001$). Pairwise comparisons with Holm-adjusted p-values showed significant differences between the peri-urban and urban sites ($p < 0.001$) and between the suburban and urban sites ($p = 0.046$), but not between the peri-urban and suburban sites ($p = 0.073$). Similarly, the proportion of cardinals tracked three months post-release differed significantly among sites (Pearson's $\chi^2 = 22.18$, df = 2, $p < 0.001$). Pairwise comparisons showed again significant differences between the peri-urban and urban sites ($p < 0.001$) and between the suburban and urban sites ($p = 0.002$), but not between the peri-urban and suburban sites ($p = 0.518$). The number of cardinals tracked at one and three months post-release, missing individuals, and confirmed mortality events are presented in Table 1.

Table 1. Number of cardinals tracked 1 and 3 months post-release, with apparent survival (48% average) and fated survival (including individuals presumed alive but outside the study area; 71% average), including known mortality events/probable causes across three sites in Montreal, Quebec, Canada between 2022 and 2025.

Site	Total number of cardinals	# tracked 1 month (apparent survival)	# tracked 3 months (apparent survival)	Missing individuals after 3 months	Confirmed mortality events and probable cause of death
Peri-urban	41	34 (83%)	28 (68%)	13	6 Predation (n = 4) Unknown (n = 2)
Suburban	31	19 (61%)	18 (58%)	13	4 Predation (n = 2) Disease (n = 1) Unknown (n = 1)
Urban	36	11 (31%)	6 (17%)	30	3 Predation (n = 1) Window collision (n = 2)

311 Core areas and home ranges of cardinals across an urban gradient

312 We estimated core areas and home ranges for individuals with ≥ 5 recorded GPS locations (36
313 individuals from the peri-urban site, 13 from the suburban site and 9 from the urban site; Figure
314 4). Between 5 and 48 points per individual were used to estimate the KDEs, with the average
315 number of points per bird being 18. Mean core area sizes (± SE) were 0.15 ± 0.04 km^2 (n = 36)
316 at the peri-urban site, 0.08 ± 0.02 km^2 (n = 13) at the suburban site, and 0.28 ± 0.11 km^2 (n = 9)
317 at the urban site. Mean total home range sizes (± SE) were 0.61 ± 0.13 km^2 (n = 36) at the peri-
318 urban site, 0.56 ± 0.11 km^2 (n = 13) at the suburban site, and 1.53 ± 0.52 km^2 (n = 9) at the
319 urban site. These sample sizes represent the number of cardinals that could be included in the
320 respective KDE analyses. Core and total home range sizes were not normally distributed
321 (Shapiro-Wilk tests: core, $W = 0.624$, $p < 0.001$; total, $W = 0.698$, $p < 0.001$), so areas were log-
322 transformed prior to analysis. Following transformation, the data met assumptions of normality
323 (Shapiro-Wilk test: 50% KDE, $W = 0.970$, $p = 0.159$; 95% KDE, $W = 0.969$, $p = 0.150$) and
324 homogeneity of variance (Levene's test: 50% KDE, $F_{2,55} = 1.07$, $p = 0.350$; 95% KDE, $F_{2,55} =$
325 0.61, $p = 0.547$). Mean core home range size (95% KDE) was highest at the urban site (Figure
326 4), but differences among sites were not statistically significant (ANOVA: $p = 0.077$). There was
327 a significant difference among the three study sites regarding total home ranges (50% KDE;
328 ANOVA: $F_{2, 55} = 3.45$, $p = 0.04$). Post-hoc Tukey's honestly significant difference tests showed
329 that the only significant difference was that the total home ranges of cardinals at our urban site
330 were larger than those at the peri-urban site (mean difference = 1.24, 95% CI [0.09, 2.38], $p =$
331 0.03); no other pairwise differences were statistically significant.

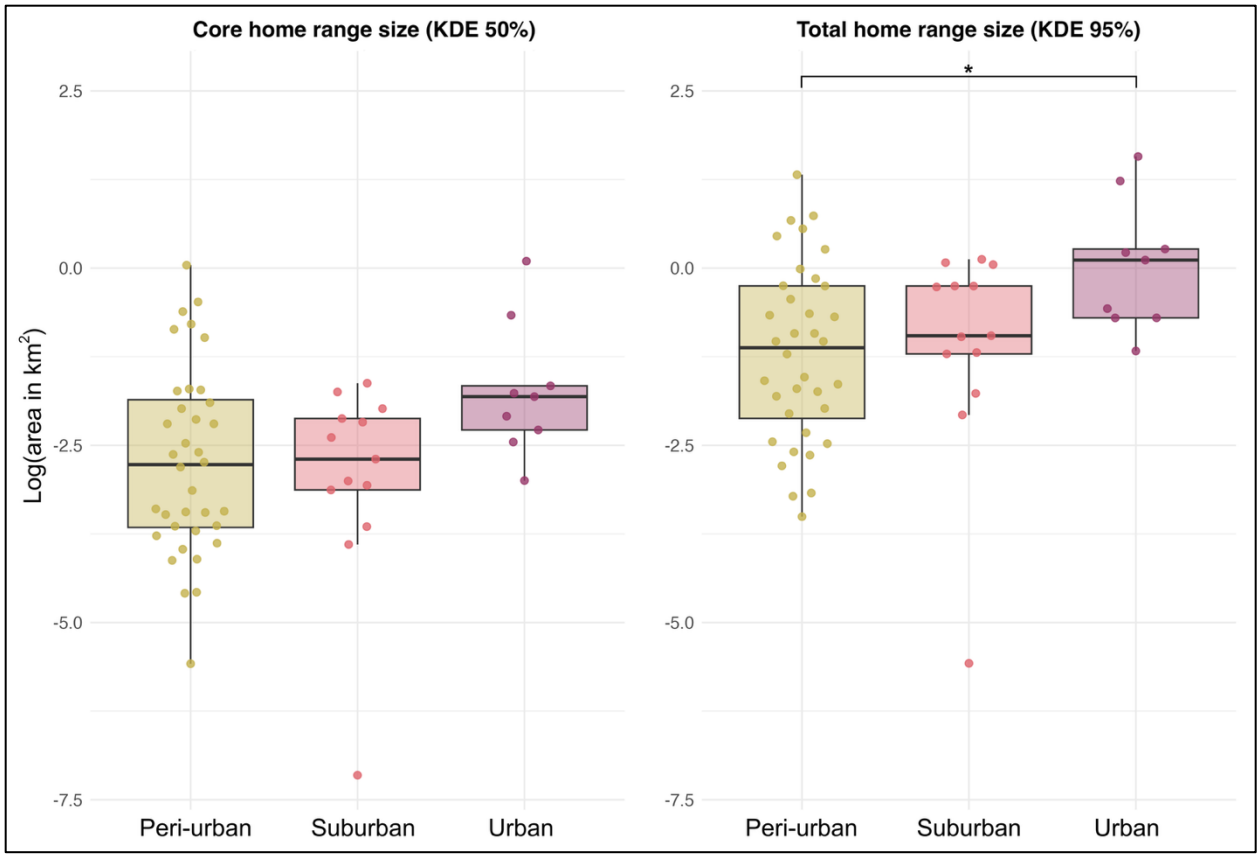


Figure 4. Boxplot comparison of the core (KDE 50%) and total (KDE 95%) home ranges of cardinals in three study sites (peri-urban, suburban and urban) in Montreal, Quebec, Canada during winters between the years 2022 and 2025. We log-transformed the area (km^2), and each data point corresponds to an individual's home range (log-transformed area).

Scale of association to habitat

We compared AICc among models fit with landscape covariates measured at four different spatial scales (25, 50, 100, and 200 m radius around resighting locations). At the peri-urban site, the 50 m scale model best explained cardinal movements (Δ AICc > 4 relative to all other scales). At both the suburban and urban sites, the 25 m scale model performed best (Δ AICc > 10 relative to all other scales). Because the landscape variables we measured at different spatial scales are inherently correlated, these results indicate the scale at which habitat structure shows the

strongest statistical association with cardinal movements, rather than reflecting a direct causal scale of effect.

Resource selection: used vs. available habitats

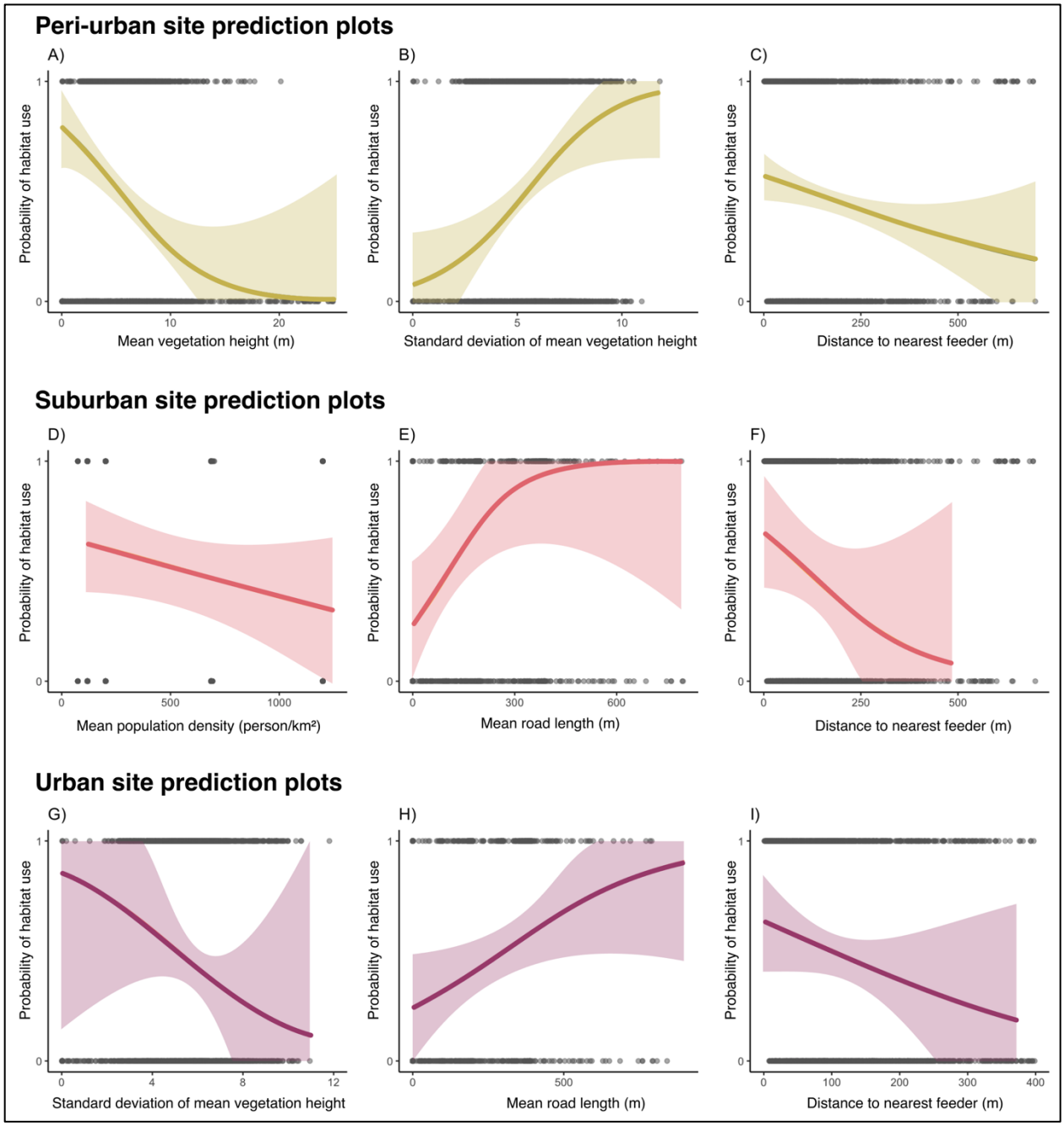
Due to differences in urbanization level and scales of effect, resource selection function models were run separately for each study site. For the peri-urban site, the global model explained the most variation seen in the data (Δ AICc > 10 relative to all other models, Supplementary Material Table S2). The full model reduced deviance by ~11.3% relative to the null model. For the suburban site, we selected the anthropogenic covariates model (human population and road density) as the most parsimonious model. Although the global model had a slightly lower AICc, the difference between the two models was very small (Δ AICc < 0.5 relative to the top model), indicating similar support for both models. Therefore, we retained the simpler anthropogenic covariates model, which reduced deviance by ~16.6% relative to the null model. Lastly, for the urban site, the global model was the best-supported candidate model (Δ AICc > 3 relative to the next best model). The global model reduced deviance by ~10.5% relative to the null model. The best model for each study site was then used to generate prediction plots on the probability of use of the habitat by cardinals according to the model variables (Table 2).

Table 2. Model parameters for the three resource selection models predicting the probability of habitat use of 56 cardinals within (25 or 50 m-radius) areas based on the average vegetation height (in meters) and its standard deviation, along with human-related environmental features such as human population density (in people/km^2), road length (in meters) and distance to the nearest bird feeder (in meters). The study site, year (winter) of sampling, and the individual's identification (ID) are included as random effects in each model.

Study Site	Variable	Coefficient estimate	Standard error	*z*	*p*
Peri-Urban	*Global model – 50 meters*				
	Mean vegetation height (m)	-1.272	0.153	-8.329	**<0.001**
	Standard deviation of the mean vegetation height	1.107	0.114	9.709	**<0.001**
	Mean population density (person/km^2)	-0.095	0.111	-0.858	0.391
	Road length (m)	0.023	0.110	0.207	0.836
	Distance to the nearest bird feeder (m)	-0.347	0.068	-5.067	**<0.001**
Suburban	*Anthropogenic covariates model – 25 meters*				
	Mean population density (person/km^2)	-0.486	0.135	-3.603	**<0.001**
	Road length (m)	1.238	0.136	9.138	**<0.001**
	Distance to the nearest bird feeder (m)	-0.427	0.128	-3.342	**<0.001**
Urban	*Global model – 25 meters*				
	Mean vegetation height (m)	0.252	0.180	1.399	0.162
	Standard deviation of the mean vegetation height	-0.458	0.191	-2.405	**0.016**
	Mean population density (person/km^2)	-0.212	0.131	-1.620	0.105
	Road length (m)	0.736	0.135	5.438	**<0.001**
	Distance to the nearest bird feeder (m)	-0.392	0.141	-2.772	**0.006**

Prediction plots of the significant fixed effects from the resource selection functions show that cardinal associations with urban habitat features varied among sites (Figure 5). In the peri-urban site, habitat use decreases as the mean vegetation height increases (Figure 5A) and the distance to the nearest bird feeder increases (Figure 5C), and increases with greater variability in vegetation height (Figure 5B). In our suburban site, cardinals' habitat use shows little relationship with population density (Figure 5D) but increases as the mean road length increases (Figure 5E). Like our peri-urban site, the use of the habitat by cardinals also decreases as the distance to the nearest feeder increases at the suburban site (Figure 5F). At the urban site, cardinals' probability of using the habitat decreases with increasing vegetation height standard deviation, although uncertainty is high (Figure 5G). The cardinals' association with habitat also increased with mean road length and decreased with increasing distance to the nearest feeders. Across sites, proximity to feeders and structural complexity of the infrastructure or the vegetation consistently influenced the probability of habitat use, yet specific relationships varied. Taken together, these results indicate that winter habitat association by cardinals varies across the urbanization gradient. While feeder availability appears to be consistently associated with a

382 higher probability of cardinals using the habitat across all sites, the habitat features associated
383 with cardinal presence differed among the peri-urban, suburban and urban sites. This suggests
384 that patterns of habitat use are largely context-dependent.



385

386 **Figure 5.** Probability of habitat use by cardinals in a 50 m radius for the suburban site and 25 m
387 radius for both the peri-urban and urban sites, as predicted by a generalized linear mixed effects

388 model. Shaded areas represent 67% confidence intervals around the predicted probabilities. The
389 model's fixed effect variables are described in Table 3, and for each model, random effects were
390 sampling years, translocation site, and bird ID.
391

392 **Discussion**

393 Our study examined winter space use, habitat associations, and apparent survival of cardinals
394 across an urban gradient. Although habitat associations differed among sites, broad patterns
395 emerged. Individuals relocated to the urban site exhibited larger home ranges and lower apparent
396 survival than their counterparts at the suburban and peri-urban sites, while associations with
397 environmental variables varied across the urbanization gradient. Proximity to bird feeders was
398 consistently associated with cardinal occurrence across all sites, and individuals at the suburban
399 and urban sites were primarily resighted in private urban green spaces (UGS) rather than larger
400 public UGS. These findings suggest that winter habitat use by cardinals may vary according to
401 local habitat conditions and resource availability, while anthropogenic food sources appear to be
402 associated with space use across our range of urban environments. Specifically, bird feeders
403 allowed the cardinal range to extend north over the past 80 years in cities, created local
404 movements from UGS to bird feeders each fall, and meant winter habitat use was tied to bird
405 feeder density. More broadly, our results indicate that private UGS may represent an important
406 component of habitat usage by cardinals during winter in southern Quebec.

407 <u>Larger home ranges at urban site and the importance of bird feeders</u>

408 We found cardinals at our most urban site had the largest total (KDE 95%) home ranges,
409 although this finding should be interpreted cautiously given the small sample size at this location
410 (n = 9) and the limited number of locations available for some of the home range estimates. In

addition, we did not have a truly rural site, which may also have had larger home ranges. Because we could not capture the full spectrum of urbanization, the relationship to territory size may not be linear (i.e., larger home ranges in rural and urban areas compared to smaller home ranges in peri-urban and suburban areas). *P. domesticus* in urban landscapes occupied smaller home ranges compared to their rural counterparts through the breeding season in Belgium (Vangestel et al. 2010). O'Donnell (2020) also suggested that birds in urban environments generally tend to have smaller home ranges. However, female cardinals venture out of their territory more often when they have supplemental feeding, compared to females that did not have access to supplemental food, leading to a trend towards larger home ranges (Humbird and Neudorf 2008). In our case, we found the opposite trend. By mapping the location of visible bird feeders, we obtained a coarse estimate of feeder density at all sites: the peri-urban site had ~13 feeders/km^2, the suburban site had ~35 feeders/$km^{2,}$ and the urban site had ~12 feeders/km^2. Although these estimates provide useful context for interpreting site-level differences, they are not representative of all the feeders available and adequately maintained for the cardinals and should therefore be interpreted cautiously. One possible explanation is that the lower feeder density at the urban site, coupled with a potentially more challenging urban matrix, may have been associated with more extensive movements between reliable resource points. Because feeder density was not directly incorporated into our home range analyses, these interpretations should be considered as being on the speculative side. Conversely, the higher feeder density observed at the suburban site might have reduced the distance needed to move among reliable food resources. However, we want to point out that our study was not designed to test this mechanism directly. Future studies that explicitly quantify feeder availability and incorporate it into movement analyses would be valuable to evaluate whether variation in feeder density in

urban areas is associated with winter home range size for birds exploiting this anthropogenic food resource.

Lower apparent survival at urban site

We found that the proportion of cardinals tracked three months post-release was similar at the peri-urban (68%) and suburban (58%) sites, but differed substantially at the urban site (17%). These differences provide insights on cardinals' persistence within monitored areas, but they cannot be directly attributed to differences in survival. The peri-urban and suburban estimates are broadly consistent with previously reported annual survival rates for cardinals elsewhere (Ohio: 50.5-78.3%, Shustack 2008; Washington, D.C.: 40-70%, Evans et al. 2015), although a direct comparison is limited by differences in study design and the nature of the data collected. The lower tracking rates experienced at the urban site may reflect an increased mortality associated with anthropogenic hazards. The urban site was the only location where window collisions were confirmed, and these accounted for 67% (2/3) of documented mortalities. Urban environments also expose birds to additional risks, including vehicle strikes and domestic cat predation (Loss et al. 2014b; Ibáñez-Álamo et al. 2020; Lopes et al. 2024). Compared with the other sites, the urban site also contained a substantially higher density of windows, including numerous multi-storey buildings, which may have increased collision risk (Loss et al. 2014a). Alternatively, some individuals may have emigrated beyond the monitored area (>18 km^2) and could not conclusively be classified as dead or alive; their subsequent fates could not be verified in the field. While confirmed mortalities demonstrate that the anthropogenic hazards contributed to losses at the urban site, the available data do not allow us to determine the relative contributions of mortality and emigration to the lower tracking rates we observed.

457 Small scale of association to habitat regardless of urbanization level

458 Cardinals were associated with habitat features at relatively fine spatial scales across all study
459 sites, with the strongest model support at 50 m in the peri-urban site and 25 m in the suburban
460 and urban sites. Although this pattern should be interpreted cautiously, given telemetry error (10-
461 15 m) and correlation among habitat variables measured at adjacent scales, it is consistent with
462 cardinals responding to fine-grained habitat features, including bird feeders, across the urban
463 gradient. More generally, our results support previous work suggesting that urban bird responses
464 can vary across spatial scales and that small habitat patches may contribute to avian habitat use
465 in cities (Dale 2018; Fidino et al. 2021; Guo et al. 2023). While our study was not designed to
466 test mechanisms underlying scale dependence, the slightly smaller scale of strongest support in
467 the suburban and urban sites is consistent with the idea that habitat configuration may become
468 increasingly important in more urbanized landscapes. Nineteen years of banding data from the
469 peri-urban site indicated a largely resident population, as the proportion of banded and unbanded
470 birds remained relatively constant throughout the year. However, a marked increase in newly
471 captured individuals during late fall (Fig. S3) suggests local movements, including post-fledging
472 dispersal and short-distance migration. Across scales, bird feeders and vegetation structure were
473 associated with habitat use from third-order habitat selection to the scale of the entire home
474 range (e.g., first order scale).
475

476 Variation of habitat associations across an urbanization gradient

477 Cardinals were associated with different environmental and anthropogenic habitat variables
478 across an urban gradient, indicating that habitat associations are different among the peri-urban,
479 suburban and urban sites. The one consistent association across study sites was the inverse

480 relationship between habitat use and distance to the closest bird feeder—which is indeed
481 apparent even at the scale of the entire range (Fig. 2; Robb et al. 2008). This is unsurprising as
482 cardinals are one of the most common species at bird feeders within its range, especially during
483 autumn and winter (Horn et al. 2014). We also found that cardinals were more likely to be found
484 in areas with higher road cover (which may reflect higher residential development) in both the
485 suburban and urban sites. This pattern is consistent with previous work showing a positive
486 association between cardinals and urbanized landscapes during the breeding season (Evans et al.
487 2015). Comparatively, cardinal associations with natural cover differed across our urbanization
488 gradient. At the urban site, cardinals avoided areas of highly contrasted vegetation structures,
489 while at the peri-urban site they were more likely to occupy an area with greater standard
490 deviation. These findings suggest that vegetation structure may be associated with winter habitat
491 use, which is consistent with a previous study in Delaware and Maryland identifying that
492 vegetation structure is a particularly important variable in urban areas, specifically for cardinals
493 during the breeding season (Duren et al. 2017). Post-fledging cardinals in urban areas strongly
494 select for habitat with a complex understory vegetation (Ausprey and Rodewald 2011). We did
495 not find evidence of the same association with vegetation structure in the peri-urban site. The
496 reason for this discrepancy remains unclear, although variation in habitat availability or habitat
497 heterogeneity among sites may be contributing factors. The importance of vegetation structure
498 and composition in birds' habitat selection is most likely scale-dependent (Deppe and Rotenberry
499 2008). Our scale of effect analysis suggests that cardinals selected wintering habitats in the peri-
500 urban location at a larger scale than the urban site. It is possible that scale-dependent differences
501 in vegetation cues contribute to this pattern, although we caution that the difference between the

scales (25 m vs. 50 m) is modest and that this interpretation remains speculative without a direct test of the vegetation composition effect.

Implications

We highlight the importance of fine-scale habitat features and resource availability in shaping the winter ecology of a resident songbird along an urban gradient. The context-dependent responses we observed underscore the importance of considering local habitat conditions and resource availability when managing UGS for resident bird species. In particular, the reliance on bird feeders and avoidance of large public UGS suggests that private UGS may play a disproportionately important role in supporting certain overwintering birds. This highlights the often-overlooked importance of private UGS in conservation, even though they are a dominant and highly diverse part of nature in cities (Goddard et al. 2010; Hutt-Taylor and Ziter 2022). While our study species is not one of conservation concern, it is culturally recognized and may serve as an important species to connect people to nature in urban areas. As urban development intensifies, integrating biodiversity-friendly practices such as preserving vertically complex vegetation and enhancing connectivity between small habitat patches may help sustain populations of resident birds that rely on UGS during the winter. Together, our results highlight how habitat association by a resident bird species is shaped by both landscape-level context and fine-scale resource availability. This multi-scale perspective demonstrates the importance of considering hierarchical habitat use in urban ecology and conservation. More broadly, our findings join a growing body of work emphasizing the ecological importance of urban landscape elements and highlight the need for inclusive UGS planning that includes both public and private green infrastructures.

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Supplemental material

Cardinals tagging procedure

We mist netted cardinals (males and females) at the McGill Bird Observatory (MBO) during October and November of the year 2022, 2023 and 2024. The birds were intercepted using 30 mm Japanese Nylon nets and placed in small cotton bags to reduce stress before handling. As part of the migration monitoring being done at the MBO, we banded all cardinals with a numbered aluminum band issued by the Canadian Bird Banding Office. The individuals were sexed, aged, weighed, and randomly assigned to one of the six possible release sites consisting of translocation sites and control sites at the capture location. In addition to the aluminum band, we fitted all cardinals with a set of heat-sealed coloured plastic bands, each with a unique combination of colours. This allowed visual identification of the individuals without having to resort to a radio tag receiver. Each bird was equipped with a Lotek NanoTag (NTQB-2) weighing 0.8 grams, using soft elastics in a figure 8 harness configuration (the total weight of the tag, harness, and glue was less than 3% of cardinals' body weight). All the radio-tags were registered as part of the Motus Wildlife Tracking System (Frei et al. 2024).

Motus tower locations

During the first winter, only 2 automated receiver towers from the Motus network were deployed. The first tower is located at the peri-urban site (4525'50.4"N 73˚56'18.6"W), while the second tower is located on top of a faculty building at McGill University's Macdonald campus (45˚24'28.3"N 73˚56'20.3"W). For the second winter, those two towers were also used for continuous data location, with the addition of a third temporary receiver tower that we installed at the suburban site (45˚24'39.3"N 73˚55'48.3"W). Finally, during the third winter, a

fourth, permanent Motus tower was installed on top of a building at Concordia University's Loyola campus location at the urban site (45˚27'32.6"N 73˚38'30.2"W).

Detailed use of gain and signal strength when tracking

The gain is a value between 0 and 100, representing the amplification of the received signal. By increasing it, the tracker increases how sensitive the receiver is to incoming radio signals. Higher gain values increase the ability to detect weaker signals but are also subject to background noise and interference. Using a lower gain reduces noise but may result in missing weak signals. In urbanized locations, the starting gain will rarely surpass 70-80 as the receiver picks up too much noise and interference. The signal is a value ranging from 0 to 200 for the SRX800 and 0 to 250 for the SRX1200. It represents the strength of the detected transmission from a radio tag, displayed in decibels (dB) on the receiver. A stronger signal indicates that the bird is closer; trackers use this information to estimate distance and direction. Trackers begin their tracking sessions with a gain of 80-70 to increase their chances of detecting nearby individuals. Once the receiver picks up a cardinals' tag, the tracker moves toward the direction with the strongest signal. Once a threshold of 150 dB (SRX800) or 200 dB (SRX1200) is attained, the trackers then lower the gain by 10 and repeat this process until they reach the tagged individual. The trackers only record a GPS point if the signal strength is 150 or 200 dB, with a gain of 40 or less. While visual confirmation was difficult to achieve due to the nature of the release habitats (high density of private property), prior testing with radio tags showed that we could assume the bird to be within 10-15 meters when the gain was ≤ 30 and the signal strength ≥ 150 dB for the SRX800 receivers (or 200 db for the SRX1200). The handheld receivers were either Lotek's SRX800 or SRX1200. They were operated by setting the frequency to 166.380 MHz (Motus tags' frequency

for the Western hemisphere) and using both the gain and the signal strength to get as close as possible to the tagged individuals. We were usually able to detect birds that were within 200 meters of our location, and on some rare occasions were even able to pick up signals of birds located as far as 800 meters away. We did encounter a significantly higher level of static noise when tracking at the urban location, yet it was not enough to prevent us from finding birds within the detection range. Another factor that could have complicated tracking efforts was the occasional aggregation of multiple birds, which can make it difficult to obtain precise location data. In most cases, no more than two birds were observed together. When larger groups were detected, the majority typically dispersed before we could approach close enough to record a GPS location. Overall, however, such aggregations were infrequent and are therefore unlikely to have significantly affected the accuracy of our tracking data.

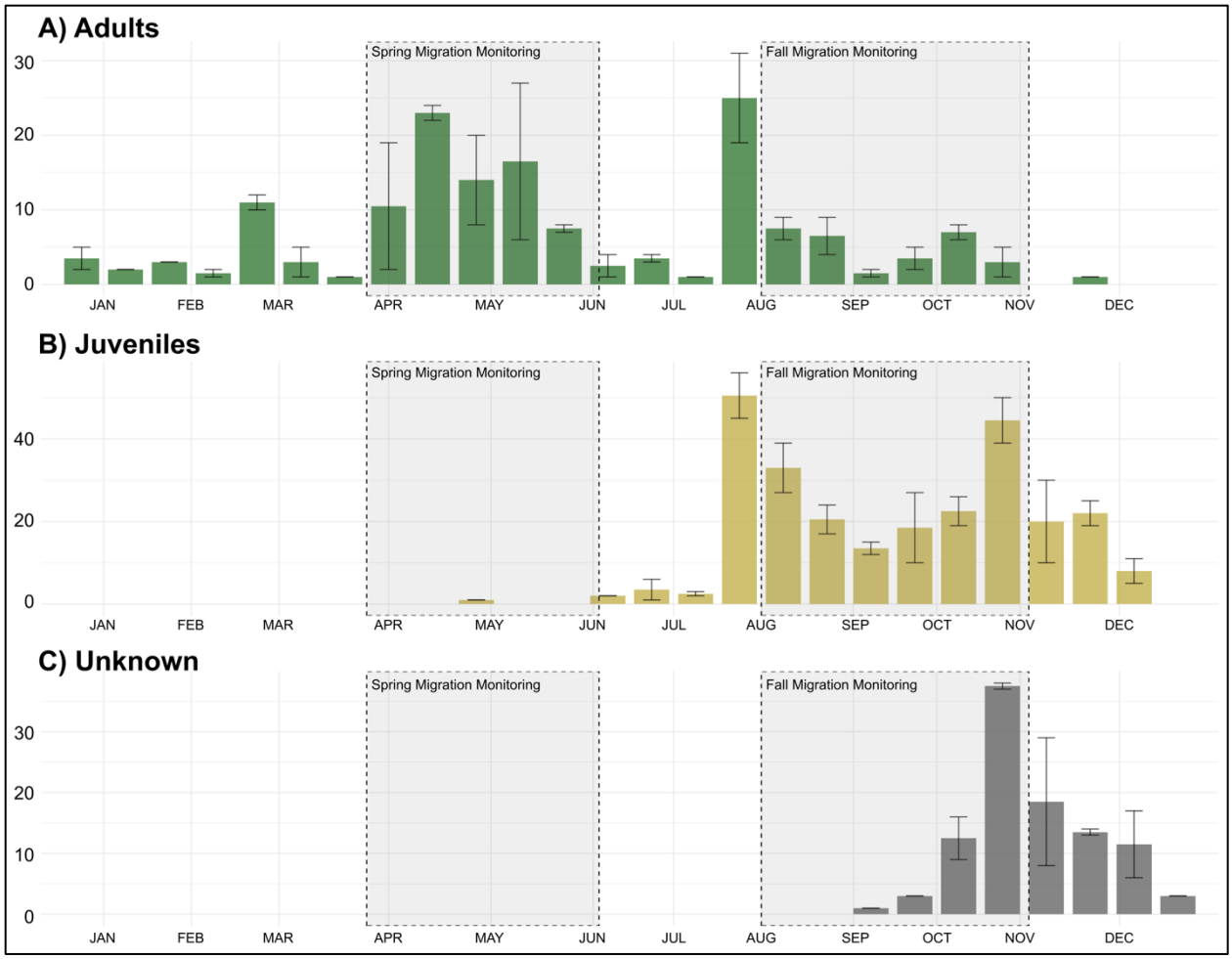


Figure S1. Mean number of cardinals captured at the McGill Bird Observatory (peri-urban site) between 2004 and 2022 in Montreal, Quebec, Canada separated by age where A) shows adult birds, B) shows juvenile birds and C) shows unknown individuals.

Table S1. Model ranking using AICc of generalized linear mixed models (GLM) with a comparing scale-of-effect of landscape variables for resighting locations (n= 57) of cardinals tracked across three sites in Montreal, Quebec, Canada. The landscape variables include the mean vegetation height, the standard deviation of the vegetation height, the mean human population density, the road length and the distance to the nearest feeder.

Study site	Model	LL	Δ AICc	Wt
Peri-urban	25 meters	-722.48	17.02	0.00
	50 meters	-713.97	**0.00**	**1.00**
	100 meters	-735.69	53.25	0.00
	200 meters	-740.60	46.96	0.00
Suburban	25 meters	-296.05	**0.00**	**1.00**
	50 meters	-338.31	84.52	0.00
	100 meters	-344.25	96.39	0.00
	200 meters	-349.92	107.73	0.00
Urban	25 meters	-211.09	**0.00**	**1.00**
	50 meters	-225.64	29.11	0.00
	100 meters	-224.02	25.85	0.00
	200 meters	-228.79	35.40	0.00

773 **Table S2.** AIC values used to determine the best fitting model to explain cardinal's habitat
774 associations.

Study site	Model	Variables	AICc
Peri-urban	Full	Mean vegetation height + sd vegetation height + mean population density + road length + distance to nearest feeder + (1|Winter) + (1|ID) + (1|Site)	**1443.696**
	Vegetation height	Mean vegetation height+ (1|Winter) + (1|ID) + (1|Site)	1587.124
	Vegetation sd	Sd vegetation height+ (1|Winter) + (1|ID) + (1|Site)	1596.077
	Human	Mean population density + road length + distance to nearest feeder+ (1|Winter) + (1|ID) + (1|Site)	1588.262
	Null	(1|Winter) + (1|ID) + (1|Site)	1616.101
Suburban	Full	Mean vegetation height + sd vegetation height + mean population density + road length + distance to nearest feeder + (1|Winter) + (1|ID) + (1|Site)	**631.7501**
	Vegetation height	Mean vegetation height+ (1|Winter) + (1|ID) + (1|Site)	728.0640
	Vegetation sd	Sd vegetation height+ (1|Winter) + (1|ID) + (1|Site)	724.0187
	Human	Mean population density + road length + distance to nearest feeder+ (1|Winter) + (1|ID) + (1|Site)	**632.1451**
	Null	(1|Winter) + (1|ID) + (1|Site)	726.1005
Urban	Full	Mean vegetation height + sd vegetation height + mean population density + road length + distance to nearest feeder + (1|Winter) + (1|ID) + (1|Site)	**444.7050**
	Vegetation height	Mean vegetation height+ (1|Winter) + (1|ID) + (1|Site)	486.6402
	Vegetation sd	Sd vegetation height+ (1|Winter) + (1|ID) + (1|Site)	485.2496
	Human	Mean population density + road length + distance to nearest feeder+ (1|Winter) + (1|ID) + (1|Site)	447.8691
	Null	(1|Winter) + (1|ID) + (1|Site)	484.8853

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